The Diffraction Grating

> SYNOPSIS
>
> Two slits give broad, faint fringes. Ten thousand slits per centimetre give maxima that are **narrow, bright and far apart** — far enough apart that each wavelength in a beam lands in its own place. The equation is the one from Note 10, unchanged. What changes is what it is good for.

BEFORE YOU START — PREDICT

The condition is still $d \sin\theta = m\lambda$ — identical to the double slit. So what does adding **thousands more slits** actually buy you, if the equation does not change?

PERFORMANCE EXPECTATIONS

1. **Compute** the slit spacing d of a grating from its ruling in lines per centimetre.
2. **Apply** $d \sin\theta = m\lambda$ to find the angle of a given wavelength in a given order.
3. **Explain** dispersion: why a grating separates white light into spectra, one per order.
4. **Locate** a line on a screen a known distance away, using $\tan\theta = y/D$.

Diffraction grating

$$d \sin\theta = m\lambda$$

where ***d* is the distance between the "lines"** — the slits. A grating is specified by how many lines are ruled per centimetre, so d is whatever is left when you divide:

WORKED, IN ONE LINE

10 000 lines per cm → $d = 1\text{ cm} / 10\,000 = \mathbf{1 \times 10^{-6}\ m}$

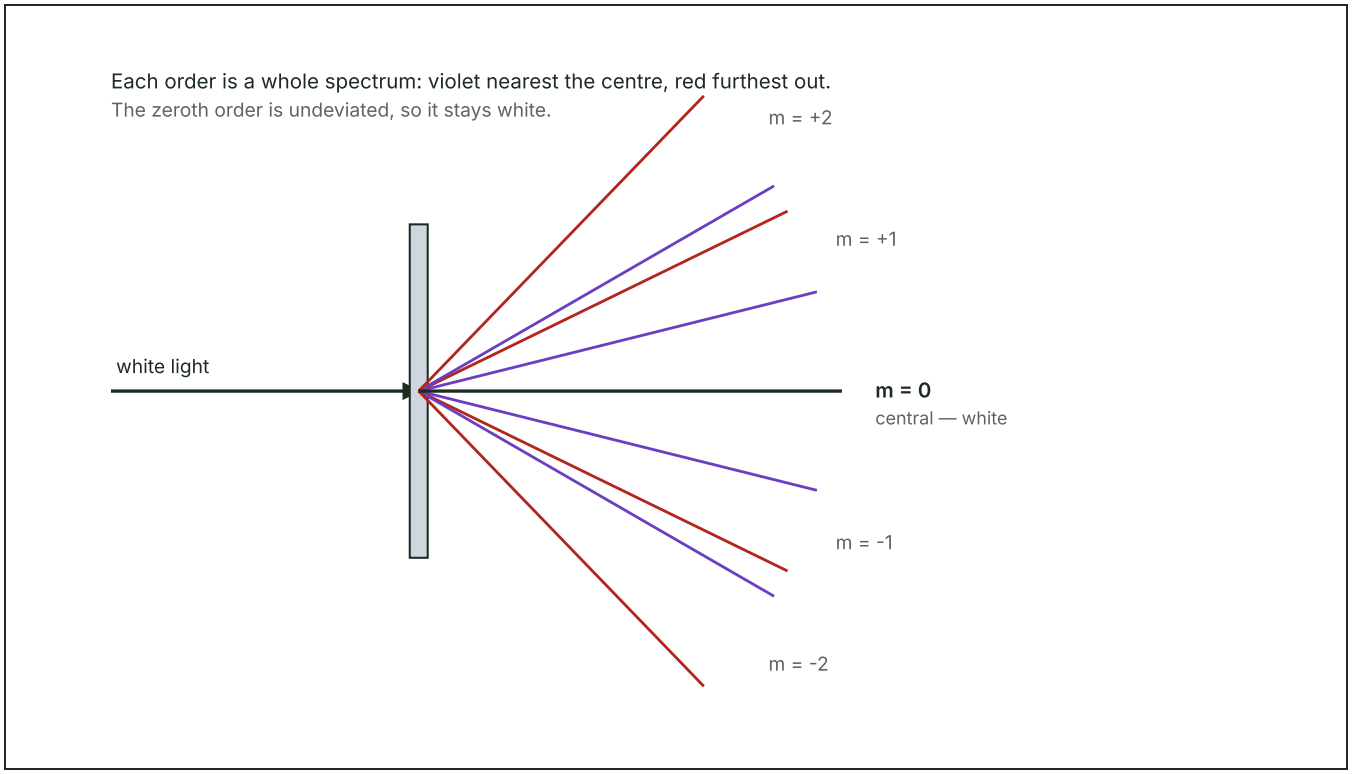


Figure 2. The orders, schematically. **No angles are marked here on purpose**: for the grating in the worked example there is no second-order red at all, because sin θ would have to be 1.52. How many orders you get depends on d.

Your job on Figure 1. The violet ray is drawn at 22.3° and its landing point marked. Work out θ_R for 760 nm, draw the red ray, and mark where it lands. Measure y_R off the figure and check it against 2 tan θ_R.

Dispersion

Rearranged, the condition reads

$$\sin \theta = m\lambda / d$$

so within one order the angle depends on λ. **Every wavelength goes somewhere different**, and white light arriving at the grating leaves as a fan of spectra — one for each order, with the zeroth order going straight through undeviated and staying white.

Worked example

Using the **10 000 lines/cm** grating above, a screen is placed **2 m** away. For **first order** (m = 1), take violet at λ_V = **380 nm** and red at λ_R = **760 nm**. Find the two angles, then y_V and y_R.

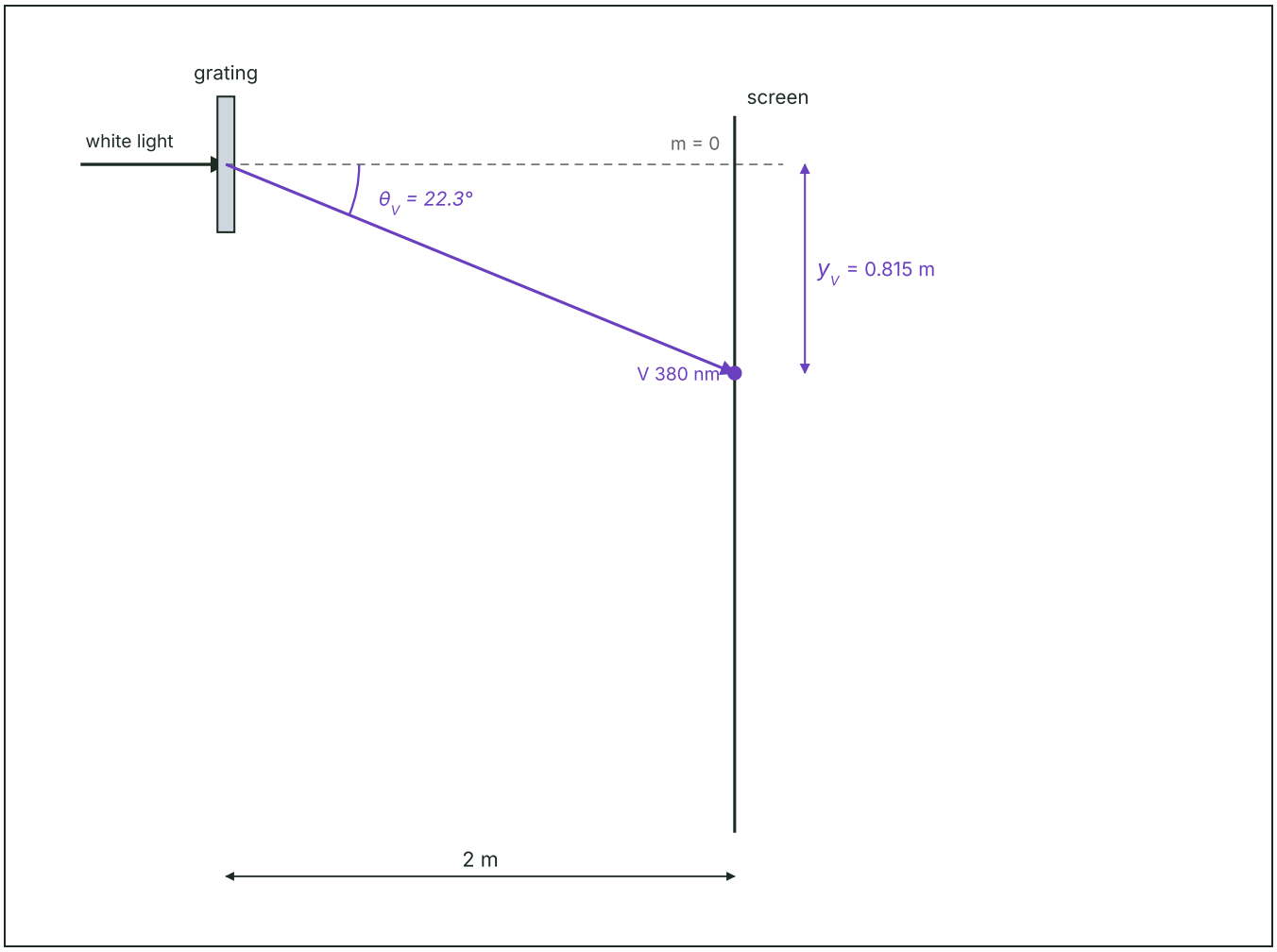


Figure 1. First order, drawn to scale. A grating does not just bend light — it bends each **wavelength** by a different amount, and by a lot. Red leaves at more than twice the angle of violet and lands nearly three times as far down the screen.

THE PUNCHLINE

$$\sin\theta = m\lambda/d \;\text{—}\; \text{one angle per wavelength}$$

Back to the prediction: the extra slits do not change *where* the maxima are. They change how **sharp** they are — with thousands of slits, light at the wrong angle is cancelled by some other slit, so each maximum narrows to a line. The equation locates them; the number of slits is what makes them worth locating.

Looking ahead

A grating separates two *wavelengths* that arrive together. **Note 16** asks the last question of the module: when two *objects* arrive together, how close can they be before diffraction makes them one blur?

2. Why is the zeroth order of a grating **white**?

○ Because the grating absorbs all colours equally there. ○ Because $m = 0$ gives $\sin\theta = 0$ for every wavelength, so none of them is deviated. ○ Because the first order has taken all the colour out. ○ It is not; the zeroth order is also a spectrum.

3. Light of 600 nm meets a grating with $d = 2.0$ µm. At what angle, in degrees, is the **second** order?

[ ]°

4. Within one order of a grating, which colour comes out at the **largest** angle?

○ Violet, because it has the shortest wavelength. ○ Red, because $\sin\theta$ grows with λ. ○ They all emerge at the same angle in a given order. ○ Green, which sits at the peak of the response.

Check Clear

(adds about 420 px to the page; re-measure and re-declare the wrapper height) --> Recording to come. Print the guided notes (link just below) and fill them in as it goes.